SOAR EDGETX BW · GUIDE

F5J M-tail model on the Pocket and GX12

Step by step: from copying the files to the radio to checking the controls and the first F5J flight. The example is the **M-tail** template: the rudder on CH6 and an elevator in two halves, each with its own servo (CH7 and CH8).

EdgeTX 2.12 | RadioMaster Pocket | RadioMaster GX12 | November 2026

Soar EdgeTX BW brings the SoarOTX soaring scripts by Jesper Frickmann to EdgeTX radios with a 128×64 black-and-white screen. For F5J it has two parts. The **timer screen** runs the flight: motor run, glide, landing, start height, the voice countdown of the working time and the score log. The **Configuration** menu lets you set up the model right on the radio, without a computer. The model template provides the flight modes, timers, logical switches, global variables, curves, mixes and channels, and the scripts read and change them.

What you need

- A RadioMaster Pocket or GX12 with EdgeTX 2.12.
- An ExpressLRS 4 receiver with at least 8 channels. The templates are set up for the radio's internal ExpressLRS module.
- A computer and a USB cable. The Pocket's microSD card can also go into a card reader.
- An altitude sensor with telemetry (the `Alt` sensor) for the start height.
- Receiver voltage telemetry (`Cels`, `RxBt`, `A1` or `A2`) for the low battery warning.
- Optionally EdgeTX Companion, to try everything in the simulator first.

Safety

- The Configuration pages write changes to the model **at once** and move the control surfaces.
- Before setup, **remove the propeller or disconnect the motor**. Do not rely on a switch position alone.
- Set up a copy of the model or work in the simulator first, then on the radio, with the model held securely.
- After setup, check the channel order, directions, endpoints and failsafe.

CONTENTS

1. Update EdgeTX and download the files
2. Copy the files to the radio
3. Create a model from the template
4. Connect the servos
5. Screens, keys and switches
6. Model setup step by step
7. Flying F5J
8. Troubleshooting
9. Checklist before the first flight

1 Update EdgeTX and download the files

1. Check the EdgeTX version: press **SYS**, then **PAGE<**: the **VERSION** page opens. You need EdgeTX 2.12 (the latest is 2.12.5). If your version is older, update the firmware as the EdgeTX manual describes (manual.edgetx.org), for example with EdgeTX Buddy.
2. Download the official SD card content for EdgeTX 2.12. From the latest 2.12 release at github.com/EdgeTX/edgetx-sdcard/releases (now v2.12.3), take `bw128x64.zip`: the Pocket and the GX12 both have a 128×64 screen.
3. Download the voice pack of the same version from github.com/EdgeTX/edgetx-sdcard-sounds/releases, for example `edgetx-sdcard-sounds-en-2.12.3.zip` for English. Without it the radio does not speak numbers: time, altitude and voltage.
4. Open the project page github.com/vitas/soar-edgetx-bw, press **Code** → **Download ZIP** and unzip it. The `dist/SDCARD` folder is not a clean SD card: it also holds models, radio settings and other files from the author's radio. Copy from it only the four items listed in section 2.

2 Copy the files to the radio

1. Connect the radio to the computer with a USB cable and choose **USB Storage (SD)**: the card shows up as a drive. The GX12 has no SD card: its internal memory shows up as the drive in the same way. The Pocket's card can also go into a card reader.
2. Back up the card (on the GX12, the internal memory), at least the `MODELS` and `RADIO` folders.
3. The official content first: unzip `bw128x64.zip` and the voice pack to the root of the card and agree to replace the files. Your models and the files you added yourself stay.
4. Then copy only these four items from `dist/SDCARD`:

What (from dist/SDCARD)	Where on the card	What for
`SCRIPTS/TELEMETRY` (the whole folder)	`/SCRIPTS/TELEMETRY`	the timer screen and the Configuration menu
`SCRIPTS/FUNCTIONS/JFutil.lua`	`/SCRIPTS/FUNCTIONS`	shared functions that the scripts load
`TEMPLATES/3.SoarEdgeTx/*.yml`	`/BACKUP`	the model templates; section 3 restores one of them as a model (create the folder if there is none)
`SOUNDS/en/*.wav`	`/SOUNDS/en`	the template's voice prompts: flight modes, launch, landing, motor off and others

5. Eject the drive safely, unplug the cable and restart the radio.

Do not copy the whole dist/SDCARD folder

Besides the package it holds models (`MODELS`), radio settings (`RADIO`), firmware and other files from the author's radio. Copied to your card, they replace your models and settings.

Updating from an earlier package

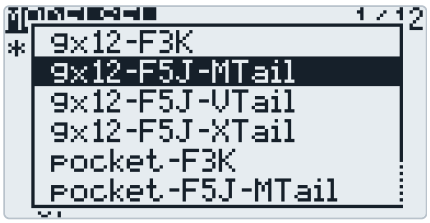
Delete these old compiled files if they are on the card: `JFXJcf.luac` and `JF5Jsk.luac` in `/SCRIPTS/TELEMETRY`, and `ARMED.luac`, `JFutil.luac` and `MENU.luac` in `/SCRIPTS/TELEMETRY/128x64`. EdgeTX runs a `.luac` file instead of its `.lua` unless the `.lua` is newer, so the old files keep the old menu (without **8. Motor idle**) or stop the screens from opening.

Radio voice in another language?
The template prompts are recorded in English and live in `/SOUNDS/en`. Copy the same `.wav` files into the folder of your radio's voice language too, for example `/SOUNDS/de`, or they may not play.

3 Create a model from the template

On black-and-white radios EdgeTX has no template list for new models: **Create model** at once makes an empty model named after its slot, such as MODEL02. The template is restored like a model backup with **Restore model**, which lists the files in the `/BACKUP` folder. EdgeTX on black-and-white radios does not read the `/TEMPLATES` folder: with the templates there and an empty or missing `/BACKUP`, **Restore model** shows "No models on SD".

1. Press MDL: the model list opens. Select an empty slot with the roller and press ENTER. The menu offers **Create model** and **Restore model**; a slot that holds a model has no **Restore model**.
2. Choose **Restore model**, select the template for your radio and press ENTER: `pocket-F5J-MTail` for the Pocket, `gx12-F5J-MTail` for the GX12. The two differ only in the model name and the switches (section 5).
3. Select the new model: ENTER on it, then **Select model**.
4. Rename the model if you like: MDL, PAGE> to **SETUP**, the **Name** field.
5. Check the RF module: the template uses the internal ExpressLRS module (CRSF, 16 channels). For ExpressLRS, first do the steps in "ExpressLRS" below. Bind the receiver.
6. Set the failsafe so that the motor stops when the signal is lost (CH3 at minimum). With ExpressLRS the failsafe is set in the receiver. Check it before the first flight (section 9).



Restore model: the templates from `/BACKUP`.

Made an empty model with Create model?
Restore the template into another empty slot and select that model: ENTER on it, **Select model**. EdgeTX does not delete the selected model, so only then delete the empty one: ENTER on it, **Delete model**, and ENTER to confirm.

ExpressLRS: update and full resolution

1. Update ExpressLRS in the radio's internal module and in the receiver to the same version 4, preferably the latest stable release. Versions 3 and 4 do not work together. Flash with ExpressLRS Configurator: choose the device, the version and the same **Binding Phrase** for the module and the receiver. Flash the module over USB (choose **USB Serial (VCP)** on the radio and the **EdgeTX Passthrough** method in Configurator) or over WiFi. Flash the receiver over WiFi: power it up without the radio, and after about a minute it opens the `ExpressLRS RX` network with the password `expresslrs`. Connect to it and open `10.0.0.1`. More at expresslrs.org.
2. Download the `elrs.lua` script and put it in `/SCRIPTS/TOOLS`. It is not tied to a firmware version. Delete the old `elrsV3.lua` and `elrsV3.luac` if they are there.
3. In the model settings (MDL, PAGE> to **SETUP**, the **Internal RF** block) set **Arm using** to "Switch" and leave the **Switch** line below it empty (---) for now. Otherwise ExpressLRS treats the model as armed while the right flap (CH5) is above center, that is with the brakes off, and the module settings must not be changed then. The **Arm using** line appears once the module runs ExpressLRS 4. The templates already have this.

4. Open SYS → **TOOLS** → **ExpressLRS** and set **Packet Rate** to "100Hz Full" and **Switch Mode** to "8ch". Of the Full modes, 100Hz has the longest range. **Switch Mode** changes only while the receiver is off: switch it off and wait for the telemetry lost alert.
5. Go back to **SETUP** and set the **Switch** line to "ON". The model is then always armed, and dynamic power, if it is on, goes to maximum when telemetry or the link is lost. To change ExpressLRS settings later, set it to "---" for that time.
6. If the receiver outputs were remapped, set them back to the defaults: output 1 to channel 1 and so on up to output 8 to channel 8.

Why Full mode

In Full mode with 8ch, ExpressLRS 4 sends channels 1–8 at full resolution, channel 5 included. In the normal modes (Hybrid, Wide) channel 5 is sent with only two positions, and channels 6–8 have low resolution. The right flap on CH5 then sits at an end, and the surfaces on CH6–CH8 move in steps. There is no need to move channels to other outputs.

4 Connect the servos

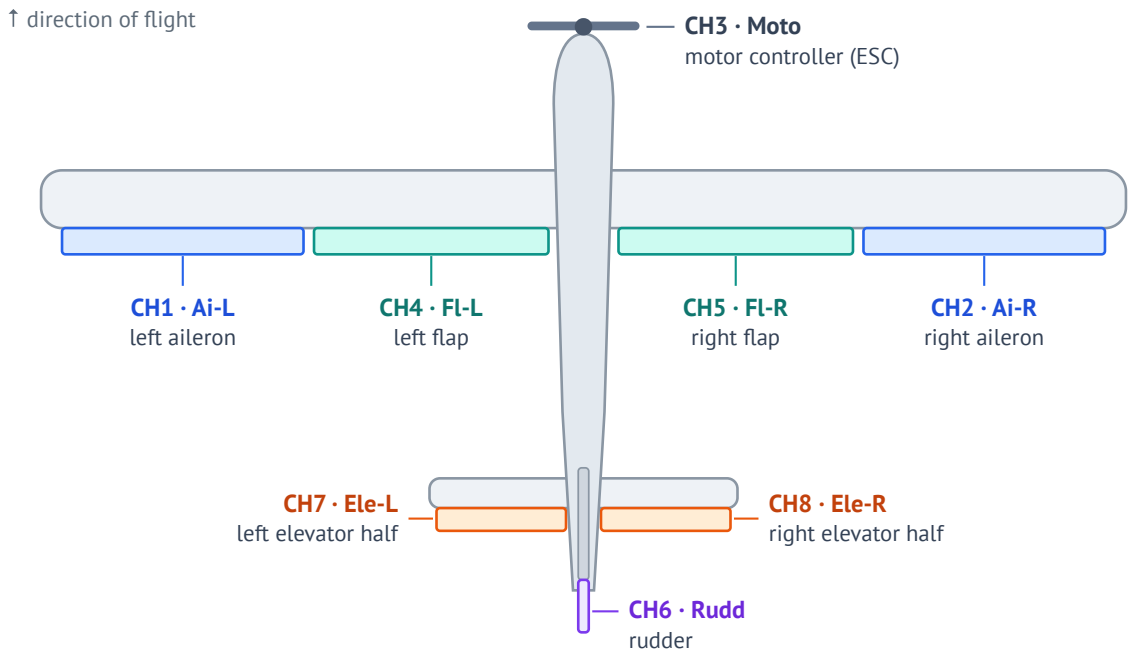


Fig. 1. Wiring for the M-tail template, top view. Next to each channel number is the output name in the model.

Channel	Name in the model	What to connect
CH1	`Ai-L`	left aileron
CH2	`Ai-R`	right aileron
CH3	`Moto`	motor controller (ESC)
CH4	`Fl-L`	left flap
CH5	`Fl-R`	right flap
CH6	`Rudd`	rudder
CH7	`Ele-L`	left elevator half
CH8	`Ele-R`	right elevator half

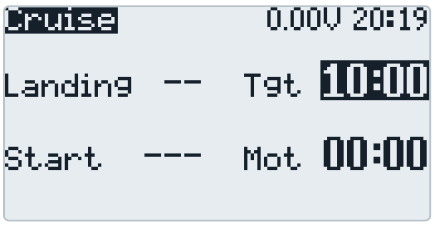
The wing and the motor are wired the same way in all F5J templates; only the tail differs. This guide describes the M-tail, and the Configuration pages work the same way with the other tails:

Channel	V-tail (`*-F5J-VTail`)	X-tail (`*-F5J-XTail`)
CH6	unused	rudder (`Rudd`)
CH7	left V-tail surface (`Vt-L`)	elevator (`Elev`)
CH8	right V-tail surface (`Vt-R`)	unused

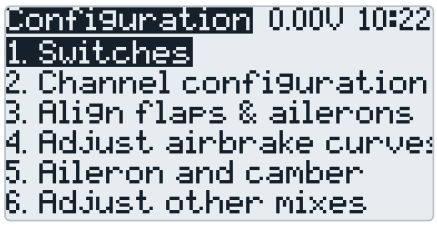
The template is neutral: centers, travels and reverses are not fitted to any particular glider; that is done in section 6. The alignment curves CV1–CV4 start linear. With the brakes off, the template puts the ailerons (CH1/CH2) at about ±37 % and the flaps (CH4/CH5) at ±67 %, so most of the flap travel stays for the brakes. The positions for normal flight are set with the last point of the airbrake curves (step 6.3).

5 Screens, keys and switches

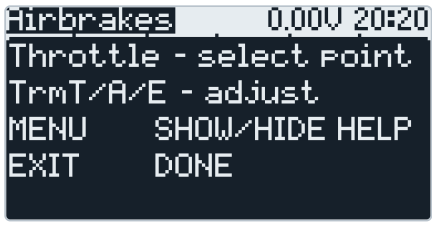
The template has two telemetry screens. **Screen 1** is the F5J timer, **screen 2** is the **Configuration** menu with eight pages. TELE opens the screens, PAGE> and PAGE< switch between them.



Screen 1: the timer, ready for a flight.



Screen 2: the Configuration menu.



Help on a page: MDL shows or hides it.

Menu item	Page title	What it sets
1. Switches	**Switches**	the switches of the template's functions (6.11)
2. Channel configuration	**Channel Config**	channel order, direction, center and endpoints
3. Align flaps & ailerons	**Alignment**	aligning the ailerons and flaps
4. Adjust airbrake curves	**Airbrakes**	airbrake (crow) curves for the flaps, the ailerons and the elevator compensation
5. Aileron and camber	**Aile & camber**	aileron travel, aileron to flap, camber to aileron
6. Adjust other mixes	**Adjust mixes**	aileron to rudder, aileron differential, brake to elevator, snapflap
7. Battery warning	**Bat Warning**	receiver battery warning level
8. Motor idle	**Motor idle**	motor power at the lowest P1 position

Keys

Key	In the Configuration menu and on its pages
Roller (turn)	select a menu item or a field, change a value; the help calls it **ROT**
ENTER (press the roller)	open a page, start or finish editing
RTN	leave a page or a field; the help calls it **EXIT**
MDL (short press)	show or hide the help of the page; the help calls it **MENU**
PAGE> / PAGE<	switch between the timer and the Configuration screens

On the timer screen the roller, ENTER and RTN have their own roles (section 7).

> **No need to save**
> Changes are written to the model at once; there is no Save button.

Switches and controls

The Pocket and GX12 templates differ only in these switches. The **1. Switches** page moves the functions to other switches (6.11).

Function	GX12	Pocket
Arm the motor	**SA** button: a press locks it in and lights it	**SA**↓
Disarm the motor (the motor stop disarms too)	**SA** button: the next press releases it	**SA**↑
Start and stop the motor	**SG** button: press and release	**SE** (spring-loaded): pull and release
Flight mode Speed / Cruise / Float	**SF**↑ / middle / **SF**↓	**SC**↑ / middle / **SC**↓
Crow on the throttle stick and landing	**SD** button, while held	**SD**↓
Crow off	**SD** released	**SD**↑
Flight time callout every 10 s	**SC**↓	**SB**↓
Motor power	**P1**	**P1**

- The motor stops when you press the start button again, when you disarm, or automatically after 30 seconds of motor run. Both radios disarm by themselves when the motor stops, so after the 10-second count the radio says the start height (section 7). To arm again on the Pocket, move **SA** out of the down position and back. On the GX12, **SA** stays in after a stop with the start button or at 30 seconds: press it twice, once to release it and once to arm. The siren and "MOTOR ARMED" confirm the arming; the **SA** light only shows that the button is in.
- **SA on the GX12** is a button. The template makes it latch in the model setup (MDL, PAGE> to **SETUP**, the **Customizable Switches** block): **SA** is set to **2POS** and is released at power-on. A press locks the button in and lights it, the next press releases it, so **SA**↓ is the button in and **SA**↑ released.
- **Throttle stick:** airbrakes (crow) while crow is on (Pocket **SD**↓, GX12 while **SD** is held); stick up deploys the brakes, stick down retracts them. With **Crow off** set to **NONE** (6.11), the stick moves the brakes all the time. In Float the throttle stick also adds thermal camber: none at the bottom, full at the top (6.4).
- **Throttle trim** down from center adds thermal camber in Cruise and Float; each of the two modes keeps its own trim. Up from center it does nothing, and in Speed it is off.

- **Elevator trim while the motor runs** sets the elevator compensation for the motor. It is stored in the Motor flight mode, and the elevator trim for the glide does not change.
- The radio announces the flight modes, crow off and motor off with the template's voice prompts.
- Free on the GX12: the **SB** and **SE** switches, the **SH** button, the **S1**/**S2** sliders, **P2** and the **SW1**–**SW6** buttons. The Pocket has no free switch.

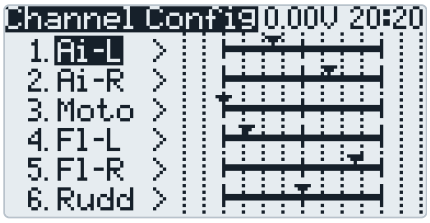
6 Model setup step by step

Open the Configuration screen (PAGE> from the timer), select a page with the roller and press ENTER; RTN returns to the menu. Do the steps in order. The motor is disconnected (except in step 6.9), and the model is held securely.

6.1. Channels and directions

2. Channel configuration

The page first warns: "Disconnect the motor!". Press ENTER only after the motor is disconnected. Each row shows the channel number, the name, the direction (`>` or `<`) and a bar with three marks (lower end, center, upper end) and a marker for the current position.



The named channels with their bars.

- Select a channel with the roller and press ENTER.
- Turn the roller to choose what to change: the number, the direction, all three marks together, the range, the lower end, the center or the upper end.
- ENTER on the direction reverses the channel. ENTER on a mark starts editing: the roller moves it, ENTER or RTN finishes.
- ENTER on the number and the roller move the channel with its mixes to another output. You need this only if the servos are not wired as in section 4.

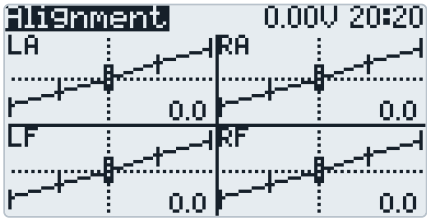
Action	Correct response
Elevator stick back	both elevator halves up
Rudder stick right	rudder right
Aileron stick right	right aileron up, left aileron down

For an M-tail, first make both elevator halves move together: reverse `Ele-L` or `Ele-R` if needed. Then adjust the center and the ends: at neutral the halves should be level, with equal travel up and down.

6.2. Wing alignment

3. Align flaps & ailerons

Aligns the four wing surfaces: **LA** and **RA** (ailerons), **LF** and **RF** (flaps). The changes are written to the curves CV1–CV4 and to the channel centers and endpoints.



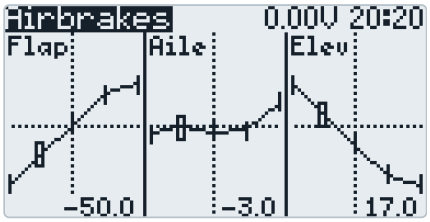
The four alignment curves.

- The throttle stick selects one of the five curve points.
- The throttle trim moves both flaps, the rudder trim aligns the left flap with the right one.
- The elevator trim moves both ailerons, the aileron trim aligns the left aileron with the right one.
- Order: the end points first, then the center, then the points between.

Starting over

ENTER on this page asks whether to reset the flap and aileron outputs. A second ENTER makes the curves CV1–CV4 straight and sets the center and the ends of the four outputs back to 0 and ±100; RTN keeps everything. If the curves and outputs do not pass the page's checks, the page opens with this question; RTN then leaves the page without changes.

6.3. Airbrakes 4. Adjust airbrake curves

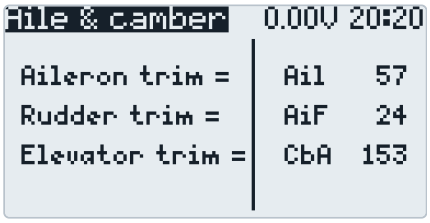


The flap, aileron and elevator curves.

Three plots: **Flap** (curve BrF), **Aile** (BrA) and **Elev** (BrE). The left end of each plot is full brakes, the right end brakes off.

- The throttle stick selects one of the five points, and the flaps and ailerons move to it.
- The throttle trim changes the flap point, the aileron trim the aileron point, and the elevator trim the elevator point.
- The last point (throttle stick down, brakes off) of **Flap** and **Aile** sets where the flaps and ailerons sit in normal flight. This is the neutral: in Speed the surfaces sit there, and Cruise and Float count their camber from it. Reflex is set through it (6.4).
- Usually, with full brakes the flaps are down and the ailerons up.
- The amount of elevator compensation is `BkE` on page 6, and the **Elev** curve shapes it along the brake travel: at −100 there is none, at 0 it equals `BkE`, and at +100 it is twice as much. The last point stays at −100, so in normal flight the elevator stays where the trim puts it.
- The elevator does not move on this page. Check the compensation after you leave the page: crow on (Pocket **SD↓**, GX12 hold **SD**), throttle stick up. If the model pitches its nose up or down as the brakes come out, note at what travel it happens and adjust that point of the **Elev** curve.
- Models made from older templates have no BrE curve: **Elev** shows N/A.

6.4. Ailerons and camber 5. Aileron and camber



The template values.

While the page is open, the trims change three values instead of moving:

Trim	Code	What it sets
Aileron	`Ail`	aileron travel
Rudder	`AiF`	aileron to flap: how much the flaps follow the ailerons
Elevator	`CbA`	camber to aileron: how much the ailerons follow the camber

The values are shared by Cruise, Landing, Speed and Float. After you leave the page, the trims work as usual again.

In flight, camber comes from the throttle trim T3 (input `CbP`), and in Float also from the throttle stick. T3 is kept separately for Cruise and Float.

Mode	Position of the flaps and ailerons
Speed (Pocket **SC**↑, GX12 **SF**↑)	No camber, and T3 does nothing: the surfaces sit at the neutral, the last points of the airbrake curves (6.3).
Cruise (SC or SF middle)	Only T3 gives camber: below center it adds camber, a little more with each click down; at center the camber is zero; above center T3 changes nothing. The template has T3 at center, so no camber.
Float (Pocket **SC**↓, GX12 **SF**↓)	Thermal camber from the throttle stick plus T3. Stick at the bottom: no thermal camber; at the top: full (32 in the template); in between: in proportion. T3 adds camber as in Cruise; the template has it at center.

In camber the ailerons follow the flaps by `CbA`. The template has 153 %, so in thermal mode the ailerons drop more than the flaps. If that is too much, lower `CbA`.

> **Thermal camber amount**
> The Pocket and the GX12 have only nine global variables, so the 32 is set directly in the `CambrX` mix of CH25 (`Camb`) and is not on the page. To change it, give that line in the mixes menu another **Weight** and the same **Offset** with a minus sign, for example 40 and −40.

Setting the camber (motor disconnected):

1. Turn the brakes off (Pocket **SD**↑, GX12 release **SD**), select Float (Pocket **SC**↓, GX12 **SF**↓) and put the throttle stick up. Open page 5: while it is open, the mode stays Float and the surfaces show the full Float camber.
2. The elevator trim changes only `CbA`: up makes the ailerons drop more, down less; the flaps do not move.
3. Leave the page, select Cruise and set the camber with T3 (down for more). In Float, T3 adds camber to the camber from the throttle stick the same way.

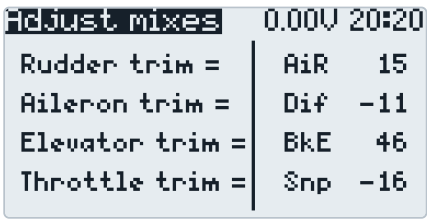
Reflex for Speed. In Speed the surfaces sit at the neutral, the last points of the airbrake curves, so reflex is set on page 4 (6.3), with the motor disconnected:

1. Open page 4 and put the throttle stick down: the last point is selected, and the flaps and ailerons sit at the neutral.
2. Raise both flaps with the throttle trim and both ailerons with the aileron trim by the reflex you want.
3. Leave the page and check in Speed (Pocket **SC**↑, GX12 **SF**↑): the surfaces sit with reflex.
4. Cruise and Float count their camber from the same neutral. In Cruise, bring the surfaces down with T3 to where you want them, for example to the old neutral; in Float, lower T3 too if needed.

6.5. Other mixes

6. Adjust other mixes

Trim	Code	What it sets
Rudder	`AiR`	aileron to rudder
Aileron	`Dif`	aileron differential
Elevator	`BkE`	elevator compensation for the airbrakes (step 6.3)
Throttle	`Snp`	snapflap: the flaps and ailerons follow the elevator stick



The template values.

As on page 5, the trims change the values instead of moving, and the values are shared by Cruise, Landing, Speed and Float. The Motor flight mode keeps its own `BkE`: there it is the elevator compensation for the motor, set with the elevator trim while the motor runs.

6.6. M-tail elevator mixes MDL menu

The M-tail template has two mixes that move the elevator halves in opposite directions, so the elevator helps the roll. Both are off in the template.

Mix (lines in CH7 and CH8)	Logical switch	In the template
`AilEle`: aileron to the elevator halves	`L46`	+20 on CH7, −20 on CH8, differential −50
`RudEle`: rudder to the elevator halves	`L48`	+20 on CH7, −20 on CH8

- The mixes work in Cruise, Landing, Speed and Float. To turn one on, choose a switch for **Ail->Ele** or **Rud->Ele** on the **1. Switches** page (6.11), or **ON** to keep the mix on all the time. The page sets the first switch of `L46` or `L48`.
- On the GX12 a free **SW1**–**SW6** button suits this. The Pocket has no free switch: there the choice is `ON` or off.
- The differential −50 halves the travel of the half that goes down. The weights are fixed in the mixes, because the Pocket and the GX12 have only nine global variables. To change the amount or the direction, edit the `AilEle` and `RudEle` lines of CH7 and CH8 in MDL → **MIXES**, and keep opposite signs on CH7 and CH8.

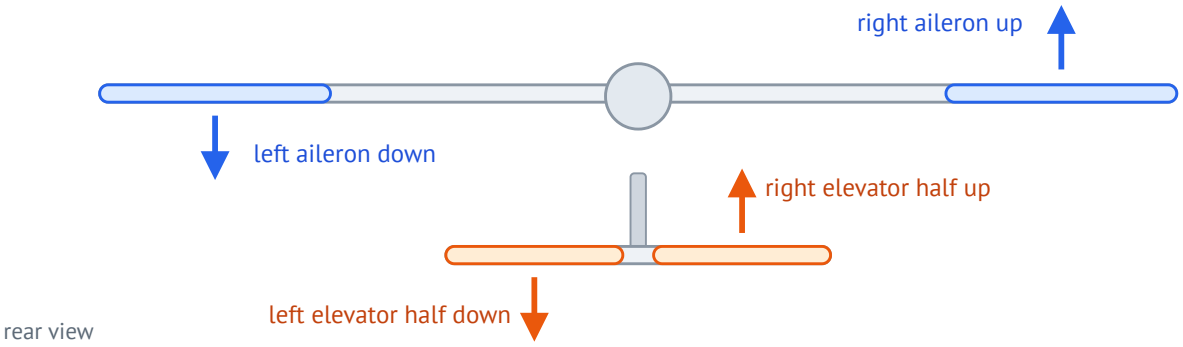


Fig. 2. Aileron stick right with the `AilEle` mix on: each elevator half works like the aileron on its side.

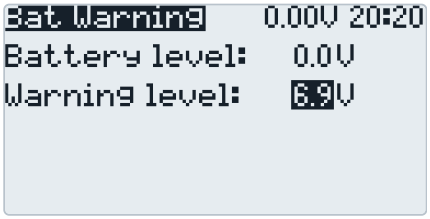
Direction check

1. Turn the mix on: **Ail->Ele** to **ON** on the **1. Switches** page.
2. Aileron stick right: the right elevator half should go up and the left one down, like the ailerons (Fig. 2). The half going down moves less.
3. If it is the other way round, swap the signs in both `AilEle` lines: −20 on CH7 and +20 on CH8.
4. `RudEle` with the same signs moves the halves with the rudder stick right the same way as the aileron stick right does. If you swapped the `AilEle` signs, swap the `RudEle` signs too.
5. Start with small values and check them in flight at a safe height.

6.7. Battery warning 7. Battery warning

Battery level shows the receiver voltage that the warning uses: the first sensor found of `Cels`, `RxBt`, `A1` and `A2`. Turn the roller to set the **Warning level** in steps of 0.1 V (1.0 to 40.0 V; the template has 6.9 V) for your receiver battery. ENTER or RTN closes the page.

Below the level the radio warns about the low battery and says the voltage, at most once every 20 seconds. While the motor runs and for 3 seconds after, the level is 10 % lower.



Without telemetry the voltage is 0.0 V.

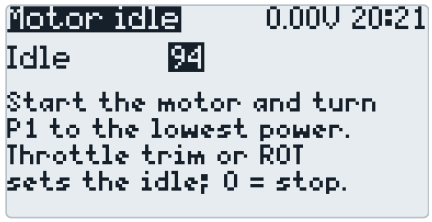
6.8. Motor check propeller off!

1. Open **2. Channel configuration** and press ENTER on the warning. The marker in the **3. Moto** row shows the motor channel CH3.
2. Without arming, the marker stays at the left end (−100) whatever P1 and the sticks do.
3. Arm (GX12 press **SA**, Pocket **SA**↓), then press and release the start button (GX12 **SG**, Pocket **SE**). The motor starts when you release it: the Motor flight mode comes on.
4. Turn P1 from end to end: the marker moves between the center mark (the lowest power) and the right end of the bar (+100, full power). Remember which end of P1 is the lowest power. The ends can be swapped (6.10).
5. Stopping: the start button again, disarming (GX12 press **SA**, Pocket **SA**↑), or automatically after 30 seconds. The marker returns to the left end.
6. Only after this check, connect the motor and fit the propeller.

6.9. Motor idle 8. Motor idle

The idle here is the motor power at the lowest P1 position. Set it by ear with the motor running. This is the only step with the motor connected and the propeller on: hold the model and do not stand in front of the propeller.

1. Open **8. Motor idle** and confirm the propeller warning with ENTER. The page itself does not start the motor.
2. Start the motor as usual (arm, then the start button) and turn P1 to the lowest power.
3. Press the throttle trim up or down, or turn the roller: each click changes the idle by 1, and the motor follows at once.
4. Stop the motor and press RTN. The idle stays in the model.



The idle after five trim clicks down from 99.

- Range 0 to 100, the template has 99. At 0 the motor stops at the lowest P1 position.
- The value is the first point of the CV13 `Pot` curve. The page scales the middle point with it, so the P1 range keeps its shape.
- While the page is open, the throttle trim does not move the camber. In a model made from an older template (without `L49` and `SF39`) the throttle trim works as usual; use the roller.

6.10. P1 and stick direction, camber on a slider optional

In the template, P1 sets the motor power, and the throttle stick has two jobs: the thermal camber in Float and the brakes while crow is on. The stick up gives full thermal camber and full brakes. Each of these can be turned around in the model's inputs (MDL → **INPUTS**) and mixes (**MIXES**), and the camber can move from the stick or T3 to a pot or slider. Check each change with the motor disconnected: the surfaces show the brakes and the camber, and the **3. Moto** row on page 2 shows the motor power (6.8).

What you want	Where	Change
The lowest power at the other end of P1	input `I4:Mot`, line `On`	curve `Mot` → `!Mot`
Full brakes and full thermal camber with the stick down	input `I5:Brk`	weight −100 → +100
Only the thermal camber with the stick down	CH25 mixes, line `CambrX`	curve `LSl` → `!LSl`

- `!Mot` is the same curve inverted, so P1 keeps its range and only its ends swap. Keep the weight −100 and do not change the `Off` line. Do not set the weight to +100 instead: full power then comes as a jump in the last 5 % of the P1 travel, and the lowest power is only at the very end. In 6.9, turn P1 to its new lowest-power end.

- The brakes and the thermal camber both come from `I5:Brk`, so its weight turns both around at once. The stick up then gives neither, and the top 5 % of the travel keep the brakes fully in, as the bottom 5 % did before. Nothing else needs changing: the brake curves follow the brake channel, with crow off (Pocket **SD↑**, GX12 **SD** released) the brakes stay in, and the brakes-in zone of the throttle crow (6.11) moves to the top with the brakes-in end.
- At power-on, the throttle warning (**T-Warning**) asks for the throttle stick down, which after this change is full brakes. To switch on with the brakes in, open MDL → **MODEL SETUP**, tick **Cust-Pos** (custom position) under **Preflight Checks** and set its value to 100: the warning then asks for the stick up.
- Leave `I8:Adj` as it is: it selects the points on the setup pages (6.2, 6.3), and they keep the stick directions given in this guide. If you want them to follow the new direction too, set the weight of `I8:Adj` to +100 as well; then read the stick positions in 6.3 and 6.4 the other way round: the last point (brakes off) is then the stick up.
- Leave the throttle reverse option in the model setup off: it turns the whole stick around at once, the setup pages included, and the directions in this guide no longer match.

Camber on a slider. On the GX12, the free **S1** or **S2** slider or **P2** can do this (section 5). If the control is missing from the source list, enable it in the radio's hardware settings (SYS → **HARDWARE**). There are two ways:

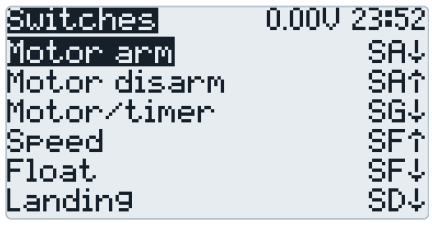
- **In addition to T3: input `I7:Cmb`.** The template keeps it for a slider; its source `MAX` gives no camber. Set the source to the slider: its +100 end (the last 5 % of the travel) gives no camber, the −100 end gives 20, on the same scale as the thermal camber (32 in the template). T3 works as before and adds to it. This camber works in every flight mode and stays while the motor runs, so in the CH25 mixes open the `Slider` line and untick FM1–FM4 (Adjust, Motor, Landing, Speed) in its flight modes: the slider then works only in Cruise and Float. Untick them in the mix line, not in the input: a switched-off input does not bring this camber to zero.
- **Instead of T3: input `I6:CbP`.** Change its source from `T3` to the slider. The slider then works like T3: below center it adds camber, at center and above it adds none. One slider position serves all modes, so Cruise and Float no longer keep their own camber. To keep Speed without camber, open the first line in the CH23 mixes (source `I6:CbP`) and untick FM1, FM3 and FM4 (Adjust, Landing, Speed) in its flight modes. As with T3, full motor power takes this camber away. T3 no longer moves the camber in flight, and the setup pages still use its buttons.

Float camber on a pot or slider. In Float the thermal camber of the throttle stick comes from the `CambrX` line in the CH25 mixes. To move it to a pot or slider, open that line and change its source from `I5:Brk` to the control; keep the weight, the offset and the curve. The control's +100 end then gives no camber and the −100 end the full 32; if it is the wrong way round, change the curve to `!LSl`. The throttle stick then moves only the brakes. On the GX12, **P2**, **S1** or **S2** is free for this; on other radios, any free pot or slider. The Pocket has no free control (P1 sets the motor power): there, untick FM5 (Float) in the flight modes of the `CambrX` line to take the camber off the stick; the throttle trim still adds camber in Float. Do one of the two when **Crow off** is **NONE** (6.11): the stick then moves the brakes all the time, and in Float it would add the camber with them.

6.11. Switches 1. Switches

The page moves the template's functions to other switches. Each row is one function, with the switch it uses now on the right. Select a row with the roller and press ENTER: the switch blinks. Turn the roller to choose another one, then press ENTER to save it or RTN to keep the old one.

The list starts with **NONE** (the function stays off) and **ON** (always on). Then come the positions with "!", which means not: **!SB↓** is on whenever **SB** is not down, so on a three-position switch it is up or middle. After them come the switch positions, on the GX12 also the **SW1**–**SW6** buttons. It wraps around: one step back from **NONE** is the last entry. The page sets the first switch of a logical switch (see the table) and stores **NONE** there as `!ON`.



The switches of the GX12 template.

Row	What it does	Logical switch	GX12	Pocket
Motor arm	arms the motor	`L41`	**SA**↓	**SA**↓
Motor disarm	disarms the motor; the motor stop disarms too	`L42`	**SA**↑	**SA**↑
Motor/timer	starts and stops the motor	`L9`	**SG**↓	**SE**↓
Speed	flight mode Speed	`L3`	**SF**↑	**SC**↑
Float	flight mode Float; with neither of the two, Cruise	`L4`	**SF**↓	**SC**↓
Landing	ends the flight once the 10-second count is over	`L7`	**SD**↓	**SD**↓
Crow off	keeps the brakes in; otherwise the throttle stick moves them	`L36`	**SD**↑	**SD**↑
Ail->Ele	the `AilEle` mix (6.6)	`L46`	**NONE**	**NONE**
Rud->Ele	the `RudEle` mix (6.6)	`L48`	**NONE**	**NONE**
Timer report	flight time callout every 10 s	`L12`	**SC**↓	**SB**↓

- On the GX12, **SD**↓ is the **SD** button held and **SD**↑ released; the same goes for **SG**. **SA** latches (section 5): **SA**↓ is the button locked in, **SA**↑ released. Two functions can share a switch, as **Landing** and **Crow off** share **SD**, and **Motor arm** and **Motor disarm** share **SA**.
- The arming is `L5`: the **Motor arm** switch turns it on when it comes on, and the **Motor disarm** switch or the motor stop turns it off. With the same switch in both rows, the arming flips each time that switch comes on, which suits a button that does not latch. The start height callout (section 7) follows the arming, not **SA**.
- A model made from an older template has no **Motor disarm** row. On the Pocket its **Motor arm** row sets `L5`, and the motor is armed while the switch is on; on the GX12 it sets `L41`, and the arming flips each time the switch comes on.

Landing at NONE. No switch ends the flight. Press the start button instead (GX12 **SG**, Pocket **SE**), as in the original SoarOTX F5J: once the 10-second count after the motor run is over, the flight timer stops and the radio says the flight time. Before that the press does nothing. It does not start the motor either: that needs a new arm.

Crow off at NONE. The throttle stick moves the brakes all the time, as in the original SoarOTX F5J. When the motor starts, the brakes lock in, so the stick position does not matter during the motor run. After the motor stops, they stay in until you move the stick to its brakes-in end, the last 5 % of the travel (the bottom in the template, the top after the change in 6.10); from there the stick moves the brakes again. While the brakes are locked and the stick is elsewhere, the radio says “Flaps up” 3 seconds after the motor stops and then every 5 seconds. In Float the stick would add the thermal camber together with the brakes, so move that camber to a pot or slider, or take it off the stick (6.10).

- The two settings are independent: either one can stay on its switch.
- A model made from an older template (without `L50`) has no lock and no reminder: with **Crow off** at **NONE** the stick just moves the brakes all the time. Restore the model from the new template to get them.

Landing and Crow off both at NONE. Only with this pair of settings does the page also turn on `L52`. The throttle stick then switches the flight mode `FM3` Landing, which takes over from Cruise, Speed and Float:

- Half a second after the stick leaves its brakes-in end (`L35`), `L53` (`L52` and not `L35`) turns Landing on, and the radio says “landing”. When the stick returns to that end, Landing goes off at once.
- Landing has the mixes of Cruise, with the same values and trims. Its elevator trim buttons set `BkE`, the elevator compensation for the airbrakes (6.5), and the elevator trim itself does not move.
- After a motor run, Landing waits like the airbrakes: it stays off until the stick has been at its brakes-in end.
- With a switch in either row, `L52` stays off and Landing never comes on, as before. A model made from an older template (without `L52`) has no Landing flight mode: restore the model from the new template to get it.

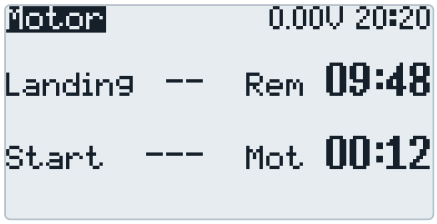
7 Flying F5J

Screen 1 shows the flight mode at the top left, the landing points (**Landing**), the start height (**Start**), the working time and the motor time (**Mot**). The working time is **Tgt** before the flight, **Rem** (remaining) in the flight and **Flt** (flown) after the landing.

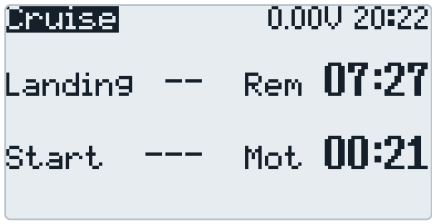
Ready → **Motor** → **Glide** → **Landed** → **Save** a motor restart goes back to Motor



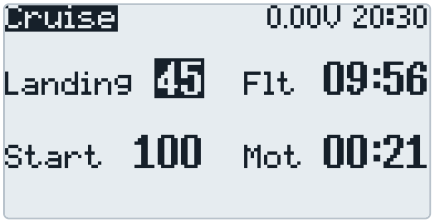
Armed: "MOTOR ARMED" flashes.



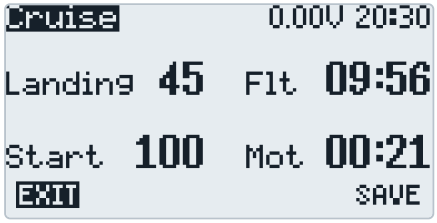
Motor run: **Rem** counts down, **Mot** up.



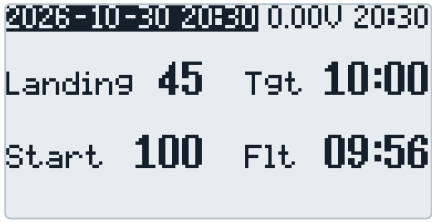
Glide.



Landed: entering the landing points.



Save: the roller selects **SAVE** or **EXIT**, ENTER confirms.



Saved scores: RTN in Ready.

1. **Ready:** the screen shows the working time **Tgt**, 10:00 by default. Turn the roller to change it in 1-minute steps (1:00 to 99:00). RTN shows the saved scores: the roller browses them, RTN returns.
2. **Arm** (GX12 press **SA**, Pocket **SA**↓): "MOTOR ARMED" flashes, the radio vibrates and sounds a siren every 5 seconds, and the altitude is reset.
3. **Start**: press and release the start button (GX12 **SG**, Pocket **SE**). P1 sets the power. The voice calls 5, 10, 15 and 20 seconds of motor run, then counts 1…9 for the seconds 21 to 29.
4. **Motor off**: the start button again, disarming (GX12 press **SA**, Pocket **SA**↑), or automatically at 30 seconds. The motor stop disarms by itself. The radio counts 10 seconds. At 10 it records the start height, the highest altitude since arming, and says it; if you arm again during the count, it says "10" instead.
5. **Glide:** the voice calls the remaining working time: every minute while more than 2 minutes remain; every 15 seconds from 2 to 1 minute; every 5 seconds from 1 minute to 10 seconds; then every second 10…1. To correct the remaining time, press ENTER: it blinks, rounded down to whole minutes. Turn the roller to change it and press ENTER to apply, or RTN to cancel.
6. **Landing:** crow on (GX12 hold **SD**, Pocket **SD**↓): the throttle stick controls the brakes. Once the 10-second count is over, the flight timer stops at once and the radio says the flight time, so use crow only for the landing. With **Landing** at **NONE** (6.11), press the start button to stop the timer; with **Crow off** at **NONE**, the stick moves the brakes without a switch.
7. **Scores:** turn the roller to set the landing points (steps of 5, 0 to 50) and press ENTER. Correct the start height (steps of 1 m, 0 to 300; 100 without an altitude sensor) and press ENTER. Correct the flight time (steps of 1 second) and press ENTER. RTN goes back one step.
8. **Save:** the selected word blinks, **EXIT** at first, so ENTER alone does not save; the roller selects **SAVE** or **EXIT**. ENTER does what is selected: **SAVE** writes the flight to `/LOGS/JF F5J Scores.csv`, **EXIT** leaves it out. The screen then returns to Ready. RTN goes back to the flight time.

Motor restart
A motor restart after the motor run (in the glide, while you enter the scores, or on the Save screen) starts a new flight at once: the working time starts again from **Tgt** and the motor time from 0. The interrupted flight is not saved. After the new motor stop come the 10-second count and the start height, the highest altitude since the restart; the altitude keeps its zero, because an arm resets it only in Ready. The motor stop disarms by itself, so the start button alone cannot restart the motor; once armed again (Pocket **SA** up and back down, GX12 **SA** released and locked in again), one press of the start button starts it. In a contest a restart scores zero under the F5J rules, and the working time runs on.

8 Troubleshooting

What you see	What to do
The model has no timer or Configuration screen	The model was made with **Create model** and is empty. Restore it from the template (section 3).
Create model offers no template	On black-and-white radios EdgeTX has no template list. Restore the template with **Restore model** on an empty slot (section 3).
Restore model shows "No models on SD" or no templates	The templates are not in `/BACKUP`, for example they were copied to `/TEMPLATES`. Copy `TEMPLATES/3.SoarEdgeTx/*.yml` to `/BACKUP` (section 2).
The menu has no **8. Motor idle** or **1. Switches**, or the screens do not open after an update	Delete the old `.luac` files (section 2) and restart the radio.
Alignment opens with "did not pass all checks"	A wing curve or output was changed outside this page, so the page cannot align it. ENTER resets the four outputs and the curves CV1–CV4; then align the wing again (step 6.2). RTN leaves them as they are.
T3 above center does nothing, no reflex	That is by design: T3 only adds camber. Reflex is the last points of the airbrake curves on page 4 (6.4).
After the motor run the brakes stay in and the radio says "Flaps up"	**Crow off** is at **NONE** (6.11): move the throttle stick to its brakes-in end, and the stick moves the brakes again.
No camber in Float	In Float the thermal camber comes from the throttle stick: at the bottom there is none. Raise the stick or lower T3 (6.4).
The help covers the page, or does not show	Press MDL: it shows or hides the help.
The motor does not start	Arm first. A motor run ends disarmed: on the GX12 press **SA** twice if it is still in (release, then arm), or move the Pocket's **SA** out of the down position and back. The motor does not start while the Alignment or Airbrakes page is open.
On the GX12, **SA** does not stay in and light up, or the motor disarms when you let go of it	**SA** does not latch: in the model setup (MDL, PAGE> to **SETUP**, the **Customizable Switches** block) set **SA** to **2POS**. A model made from an older template flips the arming at each press and needs no latch; to get the latch, restore it from the new template (section 3).
Elev shows N/A, or the throttle trim does not change the idle	The model was made from an older template, without `BrE`, `L49` and `SF39`. Use the roller for the idle, or restore the model from the new template.
The right flap (CH5) sits at an end, or CH6–CH8 move in steps	Update the module and the receiver to ExpressLRS 4 and set **Packet Rate** to "100Hz Full" and **Switch Mode** to "8ch" (section 3). Do not move channels.

What you see	What to do
The ExpressLRS script will not change a setting	The model is armed: set **Arm using** to "Switch" and the **Switch** line to "---" while you change settings, then back to "ON". **Switch Mode** changes only while the receiver is off (section 3).
The start height is always 100	There is no `Alt` sensor: discover the sensors on the model's **TELEMETRY** page.
No voice prompts for the flight modes	Copy `SOUNDS/en/*.wav` to `/SOUNDS/en`, and to the folder of your voice language (section 2).

9 Checklist before the first flight

- [ ] All control surfaces move the right way in Cruise, Speed and Float.
- [ ] The elevator halves are level at neutral and have equal travel.
- [ ] The `AilEle` and `RudEle` mixes are off, or on with small values in the right direction.
- [ ] Without arming, CH3 is at minimum; the failsafe stops the motor.
- [ ] The motor stops with the start button, when you disarm, and automatically after 30 seconds.
- [ ] Crow (Pocket **SD**↓, GX12 **SD** held; throttle stick up): flaps down, ailerons up, and the throttle stick controls the brakes smoothly.
- [ ] With **Crow off** at **NONE**: the brakes stay in while the motor runs and after it, until the throttle stick reaches its brakes-in end.
- [ ] The altitude and receiver voltage telemetry show, and the voice prompts are heard.
- [ ] A range check is done.

The project page is github.com/vitas/soar-edgetx-bw; its README describes the templates in detail. This guide describes the templates and scripts as of November 2026.